

Dr.-Ing. Sian Lun Lau

curriculum vitae

✉ sianlun@ieee.org
📁 sianlun.wordpress.com
in Sian Lun Lau
🐦 @wahlau

Education

- Nov. 2011 **Doktor der Ingenieurwissenschaften (Dr.-Ing.), *magna cum laude***, *University of Kassel*, Kassel, Germany.
- Nov. 2004 **MSc in Electrical Communication Engineering**, *University of Kassel*, Kassel, Germany.
- June 2000 **BEng with Honours in Electronic and Telecommunications Engineering**, *Universiti Malaysia Sarawak*, Kota Samarahan, Sarawak, Malaysia.

Experience

- Jan. 2021 - **Professor**, *Faculty of Engineering and Technology (FET)*, *Sunway University*,
Present Bandar Sunway, Selangor, Malaysia.
 - Execution and acquisition of research projects.
 - Teaching for undergraduate and postgraduate programmes.
 - Supervision of undergraduate and postgraduate students.
- Oct. 2025 - **Adjunct Professor**, *Vellore Institute of Technology (Chennai Campus)*, Chen-
Present nai, India.
 - Research projects and publication
 - Teaching and supervision of students.
- May 2024 - **Deputy Dean (Internationalisation)**, *School of Engineering and Technology*
Dec 2024 (*SET*), *Sunway University*, Bandar Sunway, Selangor, Malaysia.
 - Manage and implement internationalisation plans, international student mobility and recruitment for the school.
- Oct. 2021 - **Cluster Lead - Sunway DataXSight Research Cluster**, *Sunway Univer-*
Dec. 2023 *sity*, Bandar Sunway, Selangor, Malaysia.
 - Plan and manage strategies to develop research growth and impact in the area of data science
 - Coordinate interdisciplinary both internal and external collaborations.
- Mar. 2021 - **Associate Dean (Internationalization)/(International and Industry)**,
May 2024 *School of Engineering and Technology (SET)*, *Sunway University*, Bandar Sunway, Selangor, Malaysia.
 - Manage and implement internationalisation plans for the school.
 - Manage and implement international student mobility and recruitment.
 - Manage industry collaboration arrangements, working closely with Associate Deans in Research and Employability.

- Apr. 2021 - **Head of Department**, *Department of Engineering (DEN), School of Engineering and Technology, Sunway University, Bandar Sunway, Selangor, Malaysia.*
 Dec. 2023
- Management of DEN academic staff in teaching and research.
 - Planning and management of existing and new academic programmes.
- Mar. 2021 - **Associate Dean (Engagement and Employability)**, *School of Engineering and Technology (SET), Sunway University, Bandar Sunway, Selangor, Malaysia.*
 Oct. 2022
- Plan strategies and implementation to improve industry and alumni engagement for SET.
 - Manage and implement publicity and social media presence for SET.
- Jan. 2015 - **Associate Dean**, *School of Science and Technology, Sunway University, Bandar Sunway, Selangor, Malaysia.*
 Feb. 2021
- Manage academic matters at the school.
 - Assist the dean of school in day-to-day operation and management.
- Jan. 2015 - **Head of Department**, *Department of Computing and Information Systems, School of Science and Technology, Sunway University, Bandar Sunway, Selangor, Malaysia.*
 Mar. 2021
- Jan. 2015 - **Associate Professor**, *Department of Computing and Information Systems, School of Science and Technology, Sunway University, Bandar Sunway, Selangor, Malaysia.*
 Dec. 2020
- Feb. 2013 - **Senior Lecturer**, *Department of Computer Science and Networked Systems, Department of Information Systems (since Aug. 2014), Faculty of Science and Technology, Sunway University, Bandar Sunway, Selangor, Malaysia.*
 Dec. 2014
- Nov. 2011 - **Postdoctoral Researcher**, *Chair for Communication Technology, University of Kassel, Kassel, Germany.*
 Jan. 2013
- Led and contributed in EU project SEAM4US.
 - Acquisition of new research projects.
 - Supervision of student projects and theses.
- Oct. 2004 - **Researcher**, *Chair for Communication Technology, University of Kassel, Kassel, Germany.*
 Nov. 2011
- Managed and contributed in EU and national funded research projects (S4ALL (EU), MobiLife (EU), MATRIX (National)).
 - Acquisition of new research projects.
 - Supervision of student projects and theses.

Research Interests

My research interests are in the area of ubiquitous computing, specifically dealing with topics such as context awareness, user-centric computing, activity recognition, and applied machine learning. I am particularly interested in applying unobtrusive approaches in bringing future technology to everyday life in realising the vision of a smart and sustainable city. Apart from that, I also work on research related to SDG4 (Quality Education), specifically on topics like Sustainability Education and STEM and AI Education for underserved communities.

Teaching and Research Supervision Experience

Taught lectures at the School of Engineering and Technology, Sunway University:

- Bachelor degree level:
 - Programming for Engineers, Code Camp, Startup Foundry, Web and Network Analytics, Big Data Analytics in the Cloud, Concurrent Programming, Software Architecture and Design Patterns, Mobile Application Development
- Postgraduate level:
 - Smart Technology and IoT, Big Data Management, Cloud Infrastructure, Research Methodology, Mobile and Executive Computing, Enterprise Resource Planning Implementation.

Currently, I am supervising three Masters and seven PhD students:

- MSc Computer Science
 - Wong Teck Wai (Main supervisor, with Dr. Richard Wong, Sunway University)
 - Wong Yuen-Yi (Main supervisor, with Prof. David Bradley and Dr. Admad Sahban, Sunway University)
- Phd (Computing)
 - Augusta Lee Chee Sing (Main supervisor, with Assoc. Prof. Yeoh Keat Hoe, Sunway University)
 - Md Refat Khan Pathan (Main supervisor, with Prof. Mayeen Uddin Khandaker, Sunway University)
 - Muhammad Kabeer (Co-supervisor, with main supervisor Prof. Rosdiadee Nordin and Dr. Mehran Behjati, Sunway University)
 - Sivamohan A/L V Shunmugam (external, Universiti Malaya) (Co-supervisor, with main supervisor Prof. Miss Laiha Mat Kiah, University Malaya)

Students who have graduated under my supervision (only research-based degrees):

- MSc in Computer Science
 - Izzah Sakinah (Co-supervisor, with main supervisor Assoc. Prof. Dr. Tze Ying Sim, Sunway University)
 - Md Refat Khan Pathan (Main supervisor, with Dr. Peter Ho and Prof. Mayeen Uddin Khandaker, Sunway University)
 - Shanmuga Pillai A/L Murutha Muthu (Main supervisor)
 - Mihsaan Waleed (Main supervisor)
 - Cornelius Toh (Main supervisor)

- Phd
 - Lim Wei Lun (Main supervisor, with Dr. Richard Wong, Sunway University and Dr. Matthew Teow, PSB Academy Singapore)
 - Dr. Ali Taheri, PhD in Sustainable Science and Technology (co-supervisor, with Prof. Mayeen Uddin Khandaker, Sunway University)
 - Dr. Md Humayun Kabir Biswas, PhD in Computing (Main supervisor, with Assoc. Prof. Dr. Tze Ying Sim and Assoc. Prof. Dr. Saad Aslam, Sunway University)
 - Dr. Mohtashim Baqar, PhD in Computing (Main supervisor, with Prof. Mansoor Ebrahim, Iqra University, Pakistan)
 - Dr. Kazi Istiaque Ahmed, PhD in Computing (Main-supervisor, with Dr. Mohammad Tahir, University of Turku, Finland)
 - Dr. Meera Eeswaran, PhD in Business (Co-supervisor, with main supervisor Prof. Susela Devi, Sunway University Business School)
 - Dr. Mark Tee Kit Tsu, PhD in Computer Science (Co-supervisor, with main supervisor Prof. Dr. Bee Theng Lau, co-supervisor Ir. Assoc. Prof. Dr. Hudyjaya Siswoyo Jo from Swinburne University of Technology, Sarawak, Malaysia)
 - Dr. Yaqian Xu, Doktor der Ingenieurwissenschaften (Dr.-Ing.) (Co-supervisor, with Prof. Dr.-Ing. Klaus David from University of Kassel, Germany)

Served as External Examiner:

- June 2026 - Doctor of Philosophy (Viva) - Zhou Ying - *A Failure-Oriented Robustness Framework for Turbine Blade Inspection Using Learning-Based Detection and Imaging Enhancement*, Institute of Visual Informatics (IVI), Universiti Kebangsaan Malaysia
- May 2026 - Doctor of Philosophy in Computer Science (Viva) - Cheng Kam Ching - *Region-Aware Mixture-of-Expert for Fine-Grained Vehicle Damage Classification under Class Imbalance*, Tunku Abdul Rahman University of Management and Technology (TARUMT), Malaysia
- May 2026 - Master of Computer Science (Viva) - Yong Wen Lin - *Interactive Dashboard With Visual Sensing and Zero-shot Learning Capabilities*, Tunku Abdul Rahman University of Management and Technology (TARUMT), Malaysia
- April 2026 - Doctor of Philosophy (Thesis) - Agnes Jerusha J - *Detection and Classification of Fabric Defects Using Deep Learning Techniques*, Faculty of Information and Communication Engineering, Anna University
- Jan 2026 - Master in Science - Alvin Choo Ming Siang (Viva) - *Enhancing Multiclass Classification of Eye Diseases Using a Feature-Augmented Enhanced Deep Learning Approach*, Universiti Malaysia Sarawak (UNIMAS), Malaysia
- May 2025 - Doctor of Philosophy (Pre-examination) - Taneli Nyysönen, *THE PARADOX OF REALISM - Designing Virtual Reality Mechanics for Natural User Interfaces*, University of Turku, Finland

- Apr 2025 - Doctor of Philosophy (Thesis) - Maniveena C, *Deep Learning Based Attack Detection and Classification Schemes for Social IoT Data*, Noorul Islam University, India
- Nov 2024 - Doctor of Philosophy (Thesis) - Sowmiya S. - *Investigation of Human Activity Recognition using Deep Learning Approaches for IoT Wearables*, Noorul Islam University, India
- Aug 2024 - Master in Science - Dorothy Anak Martin Atok - *Extreme Gradient Boosting (XGBoost) and Random Forest (RF) Hybrid Ensemble with Bayesian Optimization (BO) in Landslide Susceptibility Mapping, Case Study: Penang Island*, Universiti Malaysia Sarawak (UNIMAS), Malaysia
- Feb 2024 - Doctor of Philosophy (Computer Science) - Menur Wahyu Pangestika - *Metaheuristic Approaches for Hospital Blood Assignment Problem*, Universiti Teknologi Mara (UiTM), Malaysia
- Aug 2023 - Phd in Information Technology (Viva) - Abdul Sattar - *A Smart Adaptive Ensemble Model to Mitigate the Issues of Concept Drift and Class Imbalance in Multi-class Data Streams*, Universiti Teknologi Petronas (UTP), Malaysia
- Dec 2022 - Doctor of Philosophy in Information and Communication Technology (Viva) - Ijaz Muhammad Khan - *An Adaptive Academic Progress Consultation Framework Based On A Novel K-Nearest Neighbors Solution for the Prediction of Students' Performance*, Universiti Tenaga National (UNITEN), Malaysia
- Oct 2022 - Master of Information Technology (Viva) - Noor Adilah Rashid - *Dual Stages Genetic Algorithm Optimized Recurrent Neural Network Long Short Term Memory*, Universiti Teknologi Petronas (UTP), Malaysia
- June 2022 - Master of Computer Science (Viva) - Samuel Ching Xin Ren - *Intelligent Manufacturing Planning System Using Dispatch Rules: A Case Study in Roofing Manufacturing Industry*, Tunku Abdul Rahman University College (TARUC), Malaysia
- Apr 2022 - Doktor der Ingenieurwissenschaften (Dr.-Ing) - Michel Morold - *Context Awareness for Smartphone-Based Cooperative VRU Collision Avoidance*, University of Kassel, Germany
- Apr 2022 - Doctor of Philosophy (Thesis) - Ninisha Nels S. - *"A Novel Security Framework for Wireless Sensor Networks*, Noorul Islam University, India
- Mar 2021 - Master of Computer Science (Viva) - Leong Chee Him - *Predictive Maintenance in Industry 4.0"*, Tunku Abdul Rahman University College (TARUC), Malaysia
- June 2020 - Doctor of Philosophy (Thesis) - D. R. Denslin Braja - *"Novel Design of Block-based Visual Cryptography Techniques for Digital Images"*, Noorul Islam University, India
- Dec 2018 - Doctor of Philosophy (Thesis) - J. Jayabharathi - *"Adaptive Computational Models to Support Reliable Service Delivery for Quality of Service Over IoT Networks"*, Annamalai University, India
- June 2017 - Doctor of Philosophy (Thesis) - Bala Murali A. - *"Efficient Dissemination of Weather Forecasting to Safeguard Farmers from Crop Failure Using Optimized Neural Network Model"*, Noorul Islam University, India
- September 2015 - Doktor der Ingenieurwissenschaften (Dr.-Ing) - Yaqian Xu - *"Autonomous Indoor Localization Using Unsupervised Wi-Fi Fingerprinting"*, University of Kassel, Germany

Served as Internal Examiner at Sunway University:

- Mar 2022 - Doctor of Philosophy (Computing) (Viva) - Chit Su Mon - *Effectiveness Of Olfactory, Audio And Haptic In Learning Things For Visually Impaired In Edutainment Environment*
- May 2019 - MSc Computer Science (Viva) - Tan Zhi Zhong - *“Zero Acceleration Algorithm with State of Motion Identifier for Position Estimation of Wheeled Robot”*
- Oct 2018 - Doctor of Philosophy (Computing) (Viva) - Yong Yoke Leng - *“Detection and Recognition of Forex Repetitive Patterns Using Machine Learning Techniques”*
- May 2018 - Doctor of Philosophy (Computing) (Viva) - Eu Kok Seng - *“Chemical Source Localisation: A Three Dimensional Technique for Quadrotors and Multi-Robot Cooperative Algorithm”*
- Jan 2017 - Doctor of Philosophy (Computing) (Viva) - Syed Aqeel Raza - *“Multi-Hop Communication in Cognitive Radio Networks: An Experimental Study”*
- April 2015 - MSc Computer Science (Viva) - Yasir Saleem - *“A Spectrum-aware Cluster-based Routing Scheme for Cognitive Radio Networks: A Simulation Study”*
- November 2013 - MSc Mobile Systems (Viva) - Law Yeong Hui - *“A Framework for Indoor Positioning System based on Android Platform”*

Professional Services

Professional Body:

- Institute of Electrical and Electronics Engineers (IEEE) - Senior member (No. 90886544)
- IEEE Computer Society Malaysia Chapter - Vice Chair/Chair-elect (2023-2026), Secretary (2021-2022), Executive committee (2017 - 2020)
- Board of Engineers Malaysia (BEM) - Graduate Engineer in Electronic (No. 172122A)
- The Institution of Engineering and Technology (IET) - Member (No. 1100850966)
- Malaysia Board of Technologists (MBOT) - Graduate Technologist in Information Technology (No. GT18030377)
- Malaysia Board of Technologists (MBOT) - Professional Technologist in Information Technology (No. PT18030411)

International Organizations:

- International Association for Convergence Science & Technology(IACST) - Malaysia Director
- Z-Inspection® Initiative - Research Member

Organizer/Chair/Co-Chair for:

- Technical Programme Co-Chair for the IEEE International Conference on Computing (ICOCO) from 2025 to 2026
- Programme Chair for the IEEE International Conference on Computing (ICOCO) from 2021 to 2024
- Organizing Team Lead, Malaysia Techlympics - Data Science Challenge for Youth category, 2022-2023
- Organizing Co-Chair for 6th International Conference on Culture Technology 2023 (ICCT)
- Publication Chairs for the 38th IEEE International Conference on Data Engineering (ICDE2022)
- Publicity Team Committee for the 2022 International Conferences of ICT for Sustainability (ICT4S)
- Technical Program Co-Chair for the 2020 IEEE Sensors Applications Symposium (SAS)
- Programme Chair for the IEEE Conference on e-Learning, e-Management and e-Services (IC3e) from 2018 to 2020
- Technical Programme Chair for the IEEE Conference on e-Learning, e-Management and e-Services (IC3e) from 2015 to 2017
- Organising Committee for the IEEE Conference on Big Data and Analytics (ICBDA) from 2017 to 2020
- Organising Committee for the IEEE Conference on Applications, Information and Network Security (AINS) from 2017 to 2020
- Organising Committee for the IEEE Conference on Open Systems (ICOS) from 2015 to 2020
- Organising Committee for the IEEE Conference on Wireless Sensors (ICWiSE) from 2015 to 2019
- Technical Program Co-Chair for the 2019 IEEE International Symposium on Haptic Audio-Visual Environments and Games (HAVE)
- International Cooperation Co-Chairs for the 33rd International Conference on Information Networking (ICOIN) 2019
- Technical Programme Chair for the 22nd International Conference on Virtual Systems & Multimedia (VSMM 2016)
- Publicity Chair for the IET International Conference on Frontiers of Communications, Networks and Applications (IET ICFCNA 2014)
- Co-chair for the AwareCast: Workshop on recent advances in behavior prediction and pro-active pervasive computing from 2013 - 2014

Editorship

- Managing Editor (2026 to present), Associate Editor (2023 to 2025), Journal of Cases on Information Technology (JCIT), IGI Global
- Editor, Blockchain and AI Technology in the Industrial Internet of Things, IGI Global (Jan 2021) (<https://www.igi-global.com/book/blockchain-technology-industrial-internet-things/256636>)
- Book editorial advisory board member for the book Assistive Technologies for Physical and Cognitive Disabilities, IGI Global (<http://www.igi-global.com/publish/call-for-papers/call-details/1117>)

Invited Talks/Guest Lectures:

- 9th December 2025, Kelana Jaya - Keynote Speaker, “*The Autonomy Paradox - When AI Systems make decisions we dont understand*”. The 4th International Conference On Data Science and Emerging Technologies 2025 (DaSET2025), Petaling Jaya, Malaysia
- 5th December 2025, Patna - Keynote Speaker, “*The Autonomy Paradox - When AI Systems make decisions we dont understand*”. First International Conference on Intelligent Machine Processes and AI-Centric Technologies (IMPACT 25), Patna, India
- 18th September 2025, Kolkata - Keynote Speaker, “*Sustainable AI for Smart Cities: Technologies that serve both People and Planet*”. 1st International Conference on Artificial Intelligence and Computing (ICAIC 2025), Jaipur, India
- 19th July 2025, Durgapur (Virtual) - Keynote Speaker, “*Sustainability and AI: Where are we heading?*”. 2025 International Conference on Computing, Intelligence, and Application (CIACON 2025), West Bengal, India
- 27th June 2025, Quezon City - Plenary Speaker, “*Development of a Cloud-based Radioluminescence Dosimetry Measurement System: Lessons learned and the way forward*”. 43rd SPP Physics Conference, Metro Manila, the Philippines
- 24th April 2025, Kolkata - Keynote Speaker, “*Pervasive Computing and Sustainable Smart Cities*”. International Conference on Computational Intelligence and Information Retrieval (CIIR 2025), Kolkata, India

- 22nd November 2024, Kolkata - *“Deep Learning for Engineering and Physical Sciences”*, International Conference on Big Data Analytics in Bioinformatics (DABCon-2024), Kolkata, India
- 13th November 2024, Kuching - *“Deep Learning for Engineering and Physical Sciences”*, IEEE Swinburne University of Technology Sarawak Student Branch, Sarawak, Malaysia
- 30th September 2024, Virtual - *“Towards Water Sustainability Through Digitisation”*, Leveraging Digital Twins for Advanced Water Supply and Sustainability Webinar, Digital Water Programme India Sub-group, Online
- 13th July 2024, Kolkata - Keynote Speaker, *“Supporting Engineering and Physical Science by Leveraging Artificial Intelligence”*. 2024 International Conference on Artificial Intelligence & Sustainable Computing (AISC), Kolkata, India
- 28th May 2024, Lappeenranta - Guest Lecture, *“Big Data and Cloud Computing”*. LUT University, Lappeenranta, Finland
- 4th August 2023, Kuching (Virtual) - Invited Speaker, *“Leveraging Data Intelligence for Sustainable Decision-Making”*. 13th International Conference on Information Technology in Asia 2023 (CITA), Kuching, Malaysia
- 17th November 2022, Bandung - Keynote Speaker, *“Towards a sustainable future through data-centric solutions: potentials and challenges”*. 7th International Conference on Computer Science and Computational Intelligence (ICCSCI), Bandung, Indonesia
- 31st March 2022, Lucknow - Invited Speaker, *“Solving Real World Problems with Big Data Analytics and Cloud”*, International Seminar on Analytic and Data Learning: Case Studies & Best Practices, Virtual Seminar, Integral University, Lucknow, India
- 7th November 2020, Kuala Lumpur - Plenary Speaker, *“30 Years Ubiquitous Computing: Where do we go from here?”*, 13th International Conference on Intelligent Robotics and Applications (ICIRA2020), Virtual Conference, Malaysia
- 23rd October 2020, Chennai - Keynote Speaker, *“30 Years Ubiquitous Computing: Where do we go from here?”*, Second International Conference on Contemporary Practices of Technology and Management for Economic Growth (ICTMEG 2020), VIT Chennai, India
- 7th July 2020, Lucknow - Invited Speaker, *“30 Years Ubiquitous Computing”*, One Week International Webinar on Artificial Intelligence & Related Technology, Integral University, Lucknow, India
- 30th June 2020, Kuala Lumpur - Plenary Speaker, *“From Internet to AI Era – How should we be ready for Industry 4.0?”*, Industry 4.0 Regional Conference 2020 (I4RC2020), Kuala Lumpur, Malaysia
- 3rd April 2019, Huizhou - Guest lecture, *“Looking beyond the Cloud (Computing)”*, Huizhou University, Huizhou, China
- 14th September 2018, Lucknow - Invited keynote speaker talk *“Keynote - Achieving a Smart and Sustainable City through Democratisation of Data”*, IEEE International Conference Computational and Characterization Techniques in Engineering & Sciences (CCTES-18), Integral University, Lucknow, India

- 11th September 2018, Chennai - Invited keynote speaker talk “*Keynote - Achieving a Smart and Sustainable City through Democratisation of Data*”, International Conference on Recent Trends in Advanced Computing - Cyber Physical Systems (ICRTAC-CPS 2018), VIT Chennai, Chennai, India
- 25th August 2017, Karachi - Invited keynote speaker talk “*Empowering and Enabling a Sustainable Smart City through IoT and IoP*”, 2017 2nd International Conference on Emerging Trends in Engineering, Sciences and Technology (ICEEST 2017), Iqra University, Karachi, Pakistan
- 17th July 2017, Kassel - Invited talk “*Bandar Sunway: Towards a Sustainable Smart City*”, ITeG Research Talk, University of Kassel, Kassel, Germany
- 15th January 2016, Noida - Invited keynote speaker talk “*From IoT to IoP - Towards Democratization of Data*”, 2016 6th International Conference - Cloud System and Big Data Engineering (Confluence), Amity University, India
- 9th January 2016, Kuala Lumpur - Invited talk “*Data Speaks Louder Than Words*”, The Star Education Fair 2016, Kuala Lumpur Convention Centre, Malaysia
- 3rd September 2015, Serdang - Invited talk “*Empowering Smart Community through Crowd-sourced Services and Open Data*”, 4th International Smart City Workshop 2015, Universiti Putra Malaysia, Malaysia
- 18th January 2015, Kuala Lumpur - Invited talk “*Speaking in Code: Career in Coding and Programming*”, The Star Education Fair 2015, Kuala Lumpur Convention Centre, Malaysia
- 5th September 2014, Bandar Sunway - Guest Lecture “*Machine Impossible*”, Lancaster-Sunway Summer School, Malaysia
- 2nd August 2014, Wernigerode - Guest Lecture “*Software Dev. @ Sunway University - Strategy, Rationale and Beyond*”, CodeCamp48 – Organized by Harz University of Applied Sciences, Germany

Panellist/Moderator:

- 24th May 2023, Virtual, Panelist, Big Data Days 2023
- 17th August 2021, Kuala Lumpur, Malaysia (Virtual) - Panelist, “*Premier Digital Tech Institutions Leadership Panel: Life Long Learning via Post-Graduate Programmes*”, Malaysia Tech Month, Online event
- 15th July 2020, Kuala Lumpur, Malaysia - Moderator, “*Panel Discussion: Industry Panel: Online or Offline Doctors?*”, #workXhealth: Work-Life Wellness in the 21st Century, Online event
- 25th June 2020, Virtual Conference - Moderator, “*Round Table 2: Reimagining and transforming learning spaces in Higher Education*”, EduTECH Virtual Asia Higher Education Conference
- 21st June 2019, Subang, Malaysia - Moderator, “*Panel Discussion: How the big tech companies are leading the digital health, and how these companies fuel digital health innovation*”, Telemedicine Conference 4.0, Grand Dorset Hotel, Subang

- 20th June 2019, Bangsar South, Malaysia - Panellist, “*CISOs discussion ? Future Cyber Threats in Malaysia*”, PIKOM Future of Cybersecurity Conference 2019, Connexion Convention and Events Centre, The Vertical
- 25th April 2019, Kuala Lumpur, Malaysia - Panellist, “*With Great Power, Comes Great Responsibility: Shaping The Socio-economic Impact of DX*”, IDC CIO Summit 2019 Kuala Lumpur, Grand Hyatt Kuala Lumpur
- 18th September 2018, Kuala Lumpur, Malaysia - Panelist, French talks - Smart Cities, Alliance Francaise de KL, Kuala Lumpur, Malaysia

Radio Interview/Media:

- 21st March 2025, Ai FM - Radio Interview, *PTPTN and Higher Education*
- 8th December 2020, Bernama Radio, *Inovasi Menerusi Pendidikan, Kerjasama, Perkongsian dan Eko-Sistem*
- 19th March 2020, Ai FM - Radio Interview, “*The Importance of Information Technology during the COVID-19 Pandemic Period*”

External Role/Membership/Services:

- Judge, Invention, Innovation & Design Exposition (IIDEX), Universiti Teknologi MARA (UiTM) Shah Alam, 2021-2025
- Co-organizer, Malaysia Machine Learning Study Jam 2025, 2025
- Chair for the Panel of Assessors, Full Accreditation for Bachelor of Computer Science, Swinburne University of Technology Sarawak Campus, 5–7 Nov 2024
- Invited Expert, appointed to ISO/TC 324 Sharing Economy by Department of Standards Malaysia (DSM), 2023-Present.
- Members of WG1 (Smart and Sustainable Cities), Malaysian Technical Standards Forum Bhd (MTSFB), 2015-Present.
- Co-organizer, Malaysia R-Conference, Malaysia R User Group (MyRUG), 2021-2024
- Mentor, SDGs in Action, York University, Canada, 2022-23
- Team Lead, Organizing committee for the Data Science Challenge for Techlympics 2022, Ministry of Science, Technology and Innovation (MOSTI), 2022-2023
- Sunway Team Advisor, Day of German Unity event, German Embassy, 2022
- Co-organizer, Malaysia Machine Learning Study Jam 2022, 2022
- Judge, Project Innovation & eXploration in CS Education and Learning (PIXEL) 2022, Universiti Sains Malaysia, 2022
- Subject Matter Expert, Development of the Industrial Skills Framework (IndSF) – Education Industry, Human Resource Development Corporation (HRD Corp), 2021
- External Auditor, Compliance Audit for Master of Engineering (Research), Swinburne University of Technology Sarawak Campus, Jan 2021
- Adjunct Research Fellow, Swinburne University of Technology Sarawak Campus - 2014 - 2019

- Judge, Young Inventors Challenge, organized by Association of Science, Technology and Innovation (ASTI), 2014-2018
- Judge, CodeCamp48 2014, organized by Harz University of Applied Sciences, 1st Aug - 3rd Aug 2014
- MyUniBus Shuttle Bus tracking web portal – System designer and maintainer, Aug 2013 to Sept 2014

Recent and Completed Externally Funded Research Projects

- August 2025 - **Co-Investigator - Bridging the Cyber Skills Gap: A Hackathon-Based Approach for Designing Cybersecurity Education**, Finnish National Agency for Education (EDUFI) Grant (51,700€ / (est.) RM 250,000.00).
- Dec 2027
- July 2025 - **Co-Investigator - Baris(Da)ta: Data-Driven Barista Training using an IoT-Enhanced Coffee Machine Platform**, Sunway University Education Research AND Innovation Grant Scheme (RM 29,150).
- June 2024
- Dec 2024 - **Co-Investigator - BEST: Blended Education for STEM Teaching**, International Science Partnerships Fund (ISPF) Grant, British Council (UK) and Academy of Sciences Malaysia (ASM) (Malaysia)((est.) £140,000/RM 800,000).
- Nov 2026
- Aug 2023 - **Co-Principal Investigator - Impact eXChange through International Hackathon for Sustainable Solutions (ImpactXChange)**, Finnish National Agency for Education (EDUFI) Grant (64,000€ / (est.) RM 325,000.00).
- Dec 2025
- Oct 2020 - **Principal Investigator - DeepSpray+: Improved Post Processing of Images from Liquid Atomization Simulations**, U.S. Army International Technology Center Pacific (\$82,000 / (est,) RM 330,660).
- Mar 2024
- Oct 2021 - **Principal Investigator - Cloud-based intelligent Measurement, Monitoring and analytics System in support of radiation medicine**, Technology Development Fund 2 (TED2), Ministry of Science, Technology and Innovation (MOSTI) (RM1,006,400.00) .
- Dec 2023
- Sep 2021 - **Co-Investigator - Development of a framework for unplugged computational thinking activities in improving the interest and skills in programming**, Fundamental Research Grant Scheme (FRGS), Ministry of Higher Education Malaysia (RM55,000) .
- Aug 2023
- Jan 2020 - **Co-Principal Investigator - SustHack - Synchronizing sustainable development actions between Finland and Malaysia - the Hackathon approach (SustHack)**, Finnish National Agency for Education (EDUFI) Grant (32,000€/ (est.) RM 153,000).
- Dec 2022
- Feb 2020 - **Co-Investigator - Research Grant for Petrosains Design Thinking Process (PDTP) Online Course Platform**, Petrosains Grant (RM15,900).
- Sept 2020

- Jan 2020 - **Co-Investigator - The impact of Computational Thinking skills on**
Dec 2020 **the Programming Skills of Novice Programmer, and Programmers**
with at least 30 hours of exposure, Sunway Internal Grant scheme
(RM15,500) .
- Feb 2019 - **Co-Investigator - ProtoPolicyAsia: empowering local communities**
Jan 2020 **and Government in Malaysia in addressing social issues in ageing and**
disabilities, UK Art & Humanities Research Council (AHRC) GCRF Highlight
Notice for International Development Grant (£85,867.32 / RM 446,510).
- Jan 2019 - **Principal Investigator - Driver Attention Management for Onroad**
Dec 2019 **Safety using IoT and Context-Awareness, Sunway Internal Grant scheme**
(RM47,500) .
- Sep 2018 - **Co-Principal Investigator (WG2) - Smart and Security Enhanced**
Aug 2020 **Transportation Ecosystem for a Future Sustainable Smart City, Mat-**
ching Grant between Sunway University and University of Malaya (RM400,000).
- Jan 2018 - **Co-Principal Investigator - Framework for Community Resilience As-**
Sep 2018 **essment and Measurement (FRAME), Lancaster University Global Chal-**
lenge Research Fund (GCRF seedcorn)(£5,000).
- Jan 2018 - **Co-Investigator - ImaginAging, Lancaster University Global Challenge**
Sep 2018 **Research Fund (GCRF seedcorn)(£5,000).**
- Jul 2014 - **Principal Investigator - Sunway Indoor GPS System, Sunway**
Jun 2017 **University- ITSSC MOU Collaboration R&D Grant (RM100,000).**
- Sep 2014 - **Principal Investigator - A crowd-source based mobile application sy-**
Dec 2016 **stem for future smart city, Sunway-Lancaster Small Grant Scheme (£7,500).**
- Jun 2014 - **Principal Investigator - Applying SAX-based Time Series Analysis**
May 2015 **to classify EEG signal using a COTS EEG device, Sunway Internal**
Grant scheme (RM31,600).
- Jul 2014 - **Co-investigator - Digital Office, MDEC Innovation Voucher (RM 15,000).**
Oct 2014
- Sep 2013 - **Co-investigator - Assistive Caretaker: A model for recognition of**
Aug 2014 **challenging behaviors for special children, Swinburne Sarawak Research**
Grant 2013 (RM5,700).
- Oct 2011 - **Project Manager/ Principal Investigator - SEAM4US (Sustainable**
Jan 2013 **Energy mAnageMent for Underground Stations), EU FP7 (Energy-**
Efficient Buildings) funded project. (Full duration of project - Oct. 2011 - Sept.
2014).
Website: <http://www.seam4us.eu/>
- Jan 2009 - **Project Manager/ Principal Investigator - MATRIX (Middleware-**
Mar 2012 **Plattform für die Realisierung internetbasierter telemedizinischer**
Dienste), German Federal Ministry of Education and Research (BMBF) fun-
ded project.
Website: <http://www.forschungsprojekt-matrix.de/de/start.htm>

- Apr 2006 - **Team member/ Co-investigator - S4ALL (Services for All)**, *EU*
 Feb 2008 *ITEA/German Federal Ministry of Education and Research (BMBF) funded project.*
 Website: <http://www.comtec.eecs.uni-kassel.de/s4all/>
- Oct 2004 - **Team member/ Co-investigator - MobiLife**, *EU CORDIS FP6 funded*
 Dec 2006 *project.*
 Website: <http://www.ist-mobilife.org>

Consultancy Projects

- Nov 2020 - **BIM – FM Platform Development for new Sunway International**
 June 2022 **School (co-investigator)**, *Sunway IMF Funded Project.*
- Nov 2020 - **Delivery and Fleet Management System (co-investigator)**, *Sunway*
 June 2022 *IMF Funded Project.*
- Oct 2021 **Fundamental of IoT**, *Training Consultancy Project with Sunway Group HR.*
- Oct 2020 - **Change Empowerment and Future Skill Mapping**, *Training Consultancy*
 Oct 2021 *Project with CIMB (Associate Trainer).*

Awards

- Jul. 2021 **Lancaster University Education Conference 2021 Award**, *The good, the bad and the “surprises” – Online Teaching and Learning feedback from the student.*
- Sept. 2019 **Jeffrey Cheah Scholar-in-Residence at Gonville and Caius College, Cambridge.**
- Jan. 2016 **Best research paper award at the 2016 6th International Conference - Cloud System and Big Data Engineering (Confluence) at Amity University, Noida, India.**
- Nov. 2014 **Special mention award in Sunway Group Innovation Challenge 2014**, *POTS for beans, Brownie Points.*
- Feb. 2014 **Silver Medal in Invention & Innovation Awards, Malaysia Technology Expo 2014 (MTEX2014)**, *Smartphone Based Auxiliary System for Drivers.*
- Nov. 2013 **2013 UK-SE Asia Knowledge Partnership-Collaborative Development Awards 2013 (together with Assoc. Prof. Dr. Yau Kok Lim, Dr Pu Chuan Chin)**, *Award: £2000 to support research visit to UK universities..*
- Nov. 2013 **Grand prize in Sunway Group Innovation Challenge 2013**, *The student-for-student (S4S) playground.*
- Nov. 2013 **2nd runner-up in Sunway Group Innovation Challenge 2013**, *The Sunway CarPool System.*
- Oct. 2013 **1st prize for the MyUniBus project in the McKinsey Youth Leadership Academy 2013**, *Technical partner for the MyUniBus project, led the development of the realtime web-based bus tracking system.*

Professional Certifications

- 2020 AWS Certified Cloud Practitioner (CLF-C01)
- 2020 Alibaba Cloud Certification Associate (ACA)
- 2017 Johns Hopkins University Data Science Specialization on Coursera (<https://coursera.org/verify/specialization/Z8DGKLRDTAQM>)
- 2016 Salesforce Sales Cloud Training and Service Cloud Essentials (SLS101, SLS201, SVC101 and SVC201)
- 2014 iTrain iOS Developer Certificate
- 2013 Huawei Certified Datacom Associate

Languages

- Human Bahasa Melayu, English, Mandarin, German
- Programming Java, Python, R, PHP
- Technology Matplotlib, Pandas, Shiny, Firebase, Android

Selected List of Publications

Journal

- [J1] K. Istiaque Ahmed, M. Tahir, Z. Jiangbin, S. Lau, M. Habaebi, and A. Ahad, "Optimizing internet of things security: Artificial neural networks algorithms performance in authentication and authorization via physical layer features," *Engineering Applications of Artificial Intelligence*, vol. 181, Oct. 2026.
- [J2] R. K. Pathan, W. L. Lim, M. U. Khandaker, C. C. Ho, and S. L. Lau, "The evolution of deep learning models for instance segmentation: a critical review and analysis of future trajectories," *International Journal of Multimedia Information Retrieval*, vol. 15, no. 3, p. 21, 2026.
- [J3] I. Ahmad, T. Sim, C. Abdul Talib, and S. Lau, "Unplugged but empowered: Investigating the impact of 3ps on computational thinking activities," *IEEE Access*, vol. 14, pp. 50340–50355, 2026. Publisher Copyright: © 2013 IEEE.
- [J4] W. L. Lim, M. Y. W. Teow, R. T. K. Wong, and S. L. Lau, "Preserving fine-grained droplet structures by spatial gaussian blurring against pooling deformation in u-net-based semantic segmentation," *IEEE Access*, vol. 14, pp. 25354–25369, 2026.
- [J5] K. Istiaque Ahmed, M. Tahir, S. Lun Lau, M. Hadi Habaebi, A. Ahad, and A. Mughees, "Trust-aware authentication and authorization for iot: A federated machine learning approach," *IEEE Internet of Things Journal*, vol. 12, no. 8, pp. 9889–9904, 2025.

- [J6] H. K. Biswas, T. Y. Sim, and S. L. Lau, “Impact of building information modelling and advanced technologies in the aec industry: A contemporary review and future directions,” *Journal of Building Engineering*, vol. 82, p. 108165, 2024.

Conferences and Workshops

- [C1] M. Kabeer, R. Nordin, M. Behjati, and S. L. Lau, “Sequence-based deep learning for handover optimization in dense urban cellular network,” in *2026 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1–6, April 2026.
- [C2] J. A. Yap, A. H. Rangkuti, S. Sen, R. Murugesan, and S. L. Lau, “Investigation on iot and cloud technology implementation for smart agriculture,” in *2025 IEEE International Conference on Computing (ICOCO)*, pp. 385–390, 2025.
- [C3] S. L. Lau, S. Aslam, A. Chakrabarti, and J. Porras, “Sustainable ai in the cloud-to-thing continuum: A bibliometric review on lightweight, small-sample, and federated learning approaches,” in *2025 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, vol. 1, pp. 1–6, 2025.
- [C4] M. O. Adisa, S. Oyediji, S. L. Lau, T. Y. Sim, and J. Porras, “Susthack: Hackathons as a model of citizen participation in innovating for sustainable urban development,” in *Citizen Participation in Sustainable Urban Development*, pp. 97 – 113, Routledge, 1 ed., 2025.
- [C5] T. W. Chin, S. L. Lau, R. T. K. Wong, L. Bravo, R. B. Koneru, and P. Khare, “Saliency-based explainable artificial intelligence (xai) for interpreting u-net semantic segmentation of synthetic liquid spray images,” in *2024 IEEE International Conference on Computing (ICOCO)*, pp. 331–336, Dec 2024.

Books/Book Chapters

- [B1] M. O. Adisa, S. Oyediji, S. L. Lau, T. Y. Sim, and J. Porras, “Susthack: Hackathons as a model of citizen participation in innovating for sustainable urban development,” in *Citizen Participation in Sustainable Urban Development*, pp. 97 – 113, Routledge, 1 ed., 2025.
- [B2] S. K. Pani, S. L. Lau, and X. Liu, eds., *Blockchain and AI Technology in the Industrial Internet of Things*. IGI Global, Jan. 2021.
- [B3] C. A. L. Lee, M. H. Yong, S. C. Lim, S. Giga, S. L. Lau, J. Johan, and E. Tseklevs, “ImaginAging Malaysia: bringing participatory speculative design to Malaysia’s exploration of healthy ageing in place,” in *Design for Global Challenges and Goals (pp. 155–169)*. (E. Tseklevs, R. Cooper, and J. Spencer, eds.), ch. 12, pp. 155–169, Taylor & Francis, 2021.
- [B4] S. L. Lau, A. Ahmed, and Z. Ruslan, “A Review for Unobtrusive COTS EEG-Based Assistive Technology,” in *Assistive Technologies for Physical and*

Cognitive Disabilities (B. T. Lau, ed.), ch. 13, pp. 262–277, Hershey, PA: IGI Global, 2015.

- [B5] S. L. Lau, *Towards a user-centric context aware system: empowering users through activity recognition using a smartphone as an unobtrusive device*. kassel university press GmbH, Jan. 2012.